Phase 2 — What Should Adaptive Therapy Optimize?

2026-09-19 · @Someone

The RL literature on adaptive cancer therapy has converged on one objective — maximise time to progression, with a bonus for holidays — while two serious alternatives, coexistence and durability, exist only in incompatible environments and have never been compared against it. This project asks whether the objective changes the policy.

Why this is open

Six papers in the corpus optimise six different things, and not one compares its objective against another. The objectives are welded to their environments, which is why the comparison has never been made.

Paper	What the reward optimises	Environment	Compares objectives?
Gallagher 2023/24	Survival time + holiday bonus	2-pop LV (Strobl)	No
Ma 2026	Same weights, "adapted from [Gallagher] and simplified"	2-pop LV, cytostatic drug	No
Giles & Newton 2026	Shannon entropy of composition (coexistence)	Moran process, fixed N	No
Weaver 2024	1 - population fitness	Empirical fitness landscapes	No
Yao 2024	Durability: regrowth after simulated withdrawal	ODE tumour-immune	No
Gluzman 2020	Cumulative dose + time penalty (not RL)	Replicator EGT	No

Giles & Newton name the gap themselves. Their introduction states that "different modeling choices encode different therapeutic objectives, ranging from tumor shrinkage and toxicity minimization to resistance management and long-term control," and positions

their entropy reward as "a complementary resistance-management objective" to Weaver's fitness minimisation. They assert complementarity and never test it. [B - the paper's own framing, verified verbatim from the PDF]

The welding is literal. Entropy over a 2-simplex is not the same quantity as entropy over Giles's 3-simplex, and a holiday bonus is undefined in a fixed-population Moran process that has no volume. Porting every objective into one environment is the work, and it is why nobody has done it. [C - inferred by comparing the six papers]

The contribution is not a new reward function. It is the first controlled comparison, which is a different and more defensible kind of claim.

Three corrections from verification

Checking the primary sources changed the build plan in three places. All three would have cost time if found in week 6 instead of week 0.

1. The proposed correctness check does not work as written. Your corpus suggests training on Gluzman's objective so the agent "should recover the HJB bang-bang policy." It will not. Gluzman solves a replicator EGT model with recovery and failure barriers, and every one of its parameters (ba = 2.5, bv = 2, c = 1, n = 4, dmax = 3, sigma = 0.01, rb = fb = 10^-1.5) is a replicator quantity with no Lotka-Volterra analogue. No published HJB solution exists for the Strobl environment. The working substitute is value iteration on a discretised (S, R) grid in your own environment, which gives a real optimality ceiling per objective rather than a borrowed one.

2. A published toxicity model now exists in the same model family. Gevertz et al., npj Systems Biology and Applications 2026;12:11, use a two-population logistic competition model where only sensitive cells are killed, plus an explicit toxicity ODE: toxicity accumulates proportionally to drug concentration and resolves linearly. It is rule-based, not RL. That makes it both a fifth objective worth including and a strong non-RL baseline nobody has benchmarked RL against. It is absent from your corpus.

3. The two papers you would copy write the drug term differently. Strobl writes the kill factor as `(1 − 2·dD·D/Dmax)`, defining `dD := 2·dD` with `dD = 1.5`. Gallagher writes `(1 − dD·D)`. These agree only if Gallagher's `dD` is Strobl's `dD = 1.5` and `D ∈ [0,1]`. A factor-of-two error here means Experiment 0 will never reproduce AT50 = 672 days, and you will spend a week debugging the agent instead of the environment.

The AT50 = 672 d and CT = 451 d figures are confirmed in the Gallagher preprint. The parameter values that produce them are not: they sit in Supplementary Information 2, which is not in the PDF on disk. Pull them from github.com/MathOnco/DRL_Personalized_AT before week 2.

The experiment

You cannot claim one objective beats another, because each objective defines its own notion of winning. Training on an objective and scoring it on its own metric is circular, and a reviewer will say so in the first paragraph.

The design that escapes this is a full cross-evaluation. Train one agent per objective, then score every agent on every metric. The scientific content lives in the off-diagonal cells.

```
flowchart LR
  E[One environment<br/>2-pop LV] --> R1[R1 survival]
  E --> R2[R2 dose+time]
  E --> R3[R3 coexistence]
  E --> R4[R4 durability]
  R1 --> M[Score every agent<br/>on every metric]
  R2 --> M
  R3 --> M
  R4 --> M
  M --> A1[Divergence:<br/>do policies differ?]
  M --> A2[Cost of the<br/>wrong objective]
  M --> A3[Instrumental<br/>convergence]
```

The matrix supports three claims, in increasing order of interest.

1. **Divergence.** Do the objectives produce different policies at all? Measure this in policy space, not outcome space: KL divergence between state-action distributions over a shared state grid, and the switching threshold N* recovered by fitting a sigmoid to P(treat) against burden. Identical thresholds would mean the objective is decorative.
2. **Cost of the wrong objective.** If a clinician cares about metric M, how much is lost by training on a different objective? That is the off-diagonal penalty, and it is the number a translational reader wants.
3. **Instrumental convergence.** Does the coexistence-trained agent achieve longer TTP than the TTP-trained agent? This is plausible rather than perverse: coexistence is instrumentally useful over long horizons, while a per-step survival bonus may be myopic. If it holds, it is the paper's headline.

Strip the RL out and the question survives: what should adaptive therapy's control target be, burden or composition or durability? West et al. pose exactly that in their open-questions survey. The agent is the instrument that answers it, because it finds a good policy for each stated objective without anyone deriving one by hand.

Environment

Use Strobl's two-population Lotka-Volterra model with turnover, in the form Gallagher writes it. Four RL papers already use it, so your numbers mean something to a reader, and its parameters are fitted to 67 real patients.

```
\begin{aligned}
\frac{dS}{dt} &= r_S\,S\left(1-\frac{S+R}{K}\right)\left(1-\hat{d}_D
D(t)\right) - d_T\,S \\[4pt]
\frac{dR}{dt} &= r_R\,R\left(1-\frac{S+R}{K}\right) - d_T\,R \\[4pt]
N(t) &= S(t) + R(t)
\end{aligned}
```

Parameter	Value	Source
r_S	0.027 /day	Strobl 2021, verified p.322 of extracted text
r_R	r_S(1 - cost), cost in [0, 0.5]	Strobl sweep; Zhang 2022 fit gives cost = 0.42
d_T (= d_S = d_R)	turnover x r_S, turnover in [0, 0.5]	Strobl; Gallagher defines turnover as d_S/r_S
K	1.0 (normalised)	Strobl non-dimensionalisation
d-hat_D	1.5	Strobl 2021, verified p.1377
n_0 = N_0/K	0.75	Strobl sweep [0.1, 1]
f_R = R_0/N_0	0.01	Strobl sweep [1e-5, 0.25]
Competition	alpha = beta = 1	Strobl; Ma 2026
Progression	N >= 1.2 N_0	Strobl, Gallagher, Ma - the field standard

Resistance is pre-existing, binary and irreversible, and the drug is cytotoxic on dividing cells via Norton-Simon. Both are assumptions to state, not facts; the corpus's Part V lists the rest.

The virtual cohort is a grid over cost of resistance and turnover, which is Strobl's and Gallagher's own parameter space. That makes your robustness heatmap directly comparable to Gallagher's Figure 5c rather than a private construction.

Two implementation traps, both cheap to avoid and expensive to find late. Check progression continuously rather than only at decision points, or a fast-growing tumour will progress and recover between observations and you will silently overstate TTP. And feed the network N/N_0 and S/K, R/K rather than raw counts, because a 10^4-scale input will not train.

The objectives

Four objectives, each ported from a named paper into the one environment. A fifth is optional if week 8 arrives early.

R1 - survival plus holiday (Gallagher 2023/24, replicated by Ma 2026). Verified verbatim from the preprint's Table 1: `+0.1` per timestep survived, `+0.05` per timestep without treatment, `-0.1` on progression, `+5` on surviving 30 years, with `gamma = 0.999`. This is the field default and your control condition.

R2 - dose and time (Gluzman 2020, negated). `r = -D(t)*dt - sigma*dt` with `sigma = 0.01`, plus a terminal bonus on reaching the time cap and a penalty on progression. Treat this as the minimise-drug-and-duration objective. Do not describe it as a check against Gluzman's HJB solution, for the reason given above.

R3 - coexistence (Giles & Newton 2026). Shannon entropy of composition, `H = -sum_i (x_i/N) log10(x_i/N)`, rewarded `+1` within 0.05% of the maximum and `-1` near the simplex boundaries.

R3 needs care, and this is the single most likely way to lose a week. Giles works on a three-population simplex where maximum entropy means three lineages coexisting. On a two-population simplex the maximum sits at S = R, which is a fifty percent resistant fraction: a naive port produces an agent that deliberately grows the resistant population, and it will look like a bug. It is not a bug, it is the objective. Handle it by running two variants: the faithful port, reported as a finding about what coexistence means when you have two

populations rather than three, and a target-fraction variant rewarding proximity to a specified resistant fraction f*, swept over {0.05, 0.1, 0.25, 0.5}. The sweep turns an awkward artefact into a result.

R4 - durability (Yao 2024). At episode end, simulate the environment untreated for a fixed horizon H and penalise the resistant population's growth over that window. Yao uses six cycles, eighteen weeks; a comparable horizon here is 180 days. This rewards leaving the tumour in a controllable evolutionary state, which is what adaptive therapy claims to do and which TTP alone does not measure.

R5 - toxicity (Gevertz 2026), optional. Add a toxicity state that accumulates proportionally to dose and resolves linearly, then penalise time spent above a threshold. This is a real toxicity model rather than a cumulative-dose proxy, and it comes with a published rule-based baseline.

Baselines and the optimality ceiling

Comparing only against MTD is not enough when tuned heuristics and computable optima both exist. Run all of these.

Baseline	Definition	Why it is here
No treatment	D = 0 always	The floor
MTD / CT	D = Dmax always	Universal comparator
AT50	Dmax until N < 0.5 N_0, then 0 until N >= N_0	The clinical protocol
AT-x sweep	Withdrawal threshold over {25, 50, 75, 90}%	The tuned heuristic ceiling - beating this is a real result
Intermittent	Fixed 30 on / 30 off	Not response-triggered
AT-PSC	Partial cycle, 13 of 30 days	Ma 2026's proposal, the current best mechanistic rule
Toxicity-feedback AT	Tumour and toxicity dual trigger	Gevertz 2026, only if R5 is included
Value iteration	Discretised (S, R) grid, per objective	The computable optimum

The last row is the one that matters most and is easiest to get wrong. Each objective has its own optimum, so value iteration must be solved separately for each, giving a column of ceilings rather than one number. The claim then becomes "the agent reached 96% of the computable optimum for its own objective," which is far more informative than "the agent beat AT50" and immunises the paper against the accusation that the agent simply trained badly.

A sanity check worth building on day one: reproduce AT50 = 672 days and CT = 451 days on Gallagher's proof-of-principle patient. Until that matches, nothing downstream is trustworthy.

Metrics

Every agent is scored on every metric, regardless of what it was trained on. That is what makes the comparison fair and the off-diagonal readable.

Metric	Definition	Which objective it favours
TTP	Days to N >= 1.2 N_0	R1
Cumulative dose	Integral of D(t), and as % of MTD	R2
Time off treatment	Fraction of episode with D = 0	R1, R2
Mean resistant fraction	Time-average of R/N	R3
Terminal resistant fraction	R/N at episode end	R3, R4
Post-withdrawal regrowth	Growth rate of R over 180 untreated days after the episode	R4
Maximum burden reached	max N(t)/N_0	none - a quality-of-life and mutation-risk proxy
Toxicity time above threshold	Days with X(t) > X_crit	R5

Report the median and the full distribution across seeds and across the cohort, never the mean alone. Wang 2024 and Eastman 2021 both show the mean hides the thing that matters, and Eastman's whole result is that RL's advantage is near-zero variance rather than a higher peak.

Two policy-space measures sit alongside these and carry the divergence claim. Fit a sigmoid to P(treat) against normalised burden and report the extracted switching threshold N* per objective, following Gallagher. Then compute pairwise KL divergence between state-action distributions over a shared state grid, following Aif's approach. Outcome metrics can coincide while policies differ, and only the second pair of measures will show it.

Train at least ten agents per objective with different seeds. Aif, Kiram and Zhu all report that single-seed results in this domain mislead, and Kiram specifically shows memoryless agents can be unstable rather than merely worse.

Experiment matrix

#	Experiment	Question it answers	Effort
E0	Implement the ODE, reproduce AT50 = 672 d and CT = 451 d	Is the environment correct?	S
E1	Baseline suite across the (cost, turnover) grid	Where is there room to improve at all?	S
E2	Value iteration per objective on a discretised (S, R) grid	What is the best achievable, per objective?	M
E3	Train R1-R4, 10 seeds each, one patient	Does each objective train stably?	M
E4	Cross-evaluation: every agent on every metric	**The core result**	M
E5	Sigmoid N* per objective, pairwise KL over the state grid	Do the policies actually differ?	S
E6	R3 target-fraction sweep over f*	Does coexistence mean anything useful with two populations?	M
E7	Cohort training, evaluate on held-out virtual patients	Does the ranking survive generalisation?	M
E8	Parameter perturbation at 15, 20, 25%	Is the ranking an artefact of one parameter set?	M

#	Experiment	Question it answers	Effort
E9	Decision-frequency sweep, train x evaluate matrix	Does objective ranking depend on monitoring cadence?	M
E10	Transfer to a second environment (cytostatic drug variant)	Does any of this survive model misspecification?	M-L

E0 through E5 are a complete and defensible project. E6 through E9 are what make it good. E10 is what makes it publishable-adjacent, and it is cheap here because the cytostatic variant is a one-line change to the drug term rather than a new simulator.

E9 deserves a note. Gallagher found that training at a lower decision frequency produces better policies and that evaluating below the training frequency fails catastrophically. If that asymmetry interacts with the objective - say, coexistence rewards tolerate sparse monitoring better than survival rewards do - that is a second publishable finding sitting inside the same experimental apparatus, at almost no extra cost.

What would kill the hypothesis

The hypothesis is that different objectives produce materially different policies with different outcome profiles. It is rejected if all four objectives converge on the same bang-bang threshold policy: extracted N* values with overlapping confidence intervals, and pairwise KL divergences no larger than the seed-to-seed divergence within a single objective.

That null result is worth publishing, which is the point of choosing this direction. It would mean the field's implicit reward-design debate is decorative in this model class, that the environment rather than the objective determines the policy, and that future effort belongs in environments rather than reward tuning. It would also strengthen the case for simple clinical rules, since if every objective lands on a threshold policy, a threshold policy is what clinicians should use.

The design must be able to produce that answer, so the noise floor has to be measured rather than assumed. Within-objective seed spread is the null distribution, and between-objective differences are only real if they exceed it. This is why ten seeds per objective is structural rather than good practice.

Three failure modes will masquerade as the null and must be excluded before claiming it.

- **The agents did not train.** Excluded by E2: an agent far below its own objective's computable optimum has not converged, and its policy says nothing about the objective.
- **Reward scales are mismatched.** An objective whose rewards are an order of magnitude smaller trains more slowly at fixed hyperparameters. Normalise each reward to a comparable per-step scale and report the learning curves.
- **The environment is wrong.** Excluded by E0. If AT50 does not give 672 days, stop and fix it before interpreting anything.

Do not design a comparison where one objective is guaranteed to win. If R1 wins on TTP, R3 on resistant fraction and R4 on regrowth, each on its own metric and nowhere else, that is a tidy but uninteresting outcome and should be reported plainly as such.

Twelve-week timeline

Starting the week of 22 September, this lands on 14 December. Weeks 3 and 4 are deliberately generous because nobody on the team has debugged an RL agent before, and that is where first projects stall.

Week	Dates	Work	Deliverable
1	Sep 22-28	Environment, baselines, E0	AT50 = 672 d reproduced
2	Sep 29-Oct 5	E1 baseline suite, start E2 value iteration	Heatmap of baseline TTP over (cost, turnover)
3-4	Oct 6-19	PPO harness, R1 only, 10 seeds, E3	One objective training stably
5	Oct 20-26	Implement R2, R3, R4; E6 f* sweep	Four objectives training
6-7	Oct 27-Nov 9	E4 cross-evaluation	**The core result matrix**
8	Nov 10-16	E5 policy divergence, sigmoid and KL	Divergence figure
9	Nov 17-23	E7 cohort, E8 perturbation	Robustness tables
10	Nov 24-30	E9 frequency sweep, E10 transfer if time	Generalisation results

Week	Dates	Work	Deliverable
11	Dec 1-7	Figures, statistics, effect sizes	Figure set frozen
12	Dec 8-14	Writing, reproducibility package	Manuscript draft and clean repo

The decision point is the end of week 4. If R1 is not training stably by then, drop to the minimum viable version rather than pushing the schedule: a late pivot is what turns a good course project into an unfinished one.

Compute is not a constraint. The environment is two coupled ODEs, PPO with a small MLP on a 10^5 to 10^6 step budget trains on a laptop CPU in hours, and Gallagher deliberately chose A3C to avoid needing a GPU at all. Value iteration on a coarse (S, R) grid is seconds. The binding constraint is your time, not the hardware.

Reviewer attacks

Criticism	What answers it
"This is just reward tuning."	The claim is about divergence between published objectives, not a new reward. No objective here is ours.
"Objectives are incommensurable, so the comparison is meaningless."	That is why the result is the cross-evaluation matrix, not a ranking. The off-diagonal is the finding.
"Your agents simply trained badly on the objectives that lost."	E2 gives a per-objective computable optimum. Every agent is reported as a fraction of its own ceiling.
"The result is an artefact of one environment."	E10 transfers to a cytostatic-drug variant. Aif's finding that rankings do not transfer is cited as the reason it is there.
"One parameter set."	E7 cohort training and E8 perturbation at 15, 20 and 25%.
"Single seed."	Ten seeds per objective, with within-objective spread as the declared noise floor.

Criticism	What answers it
"You only beat MTD."	Tuned AT-x sweep, AT-PSC, Gevertz toxicity-feedback rule, and the value-iteration optimum.
"Entropy over two populations targets 50% resistance, which is absurd."	Stated up front as a property of the port, and generalised by the f* sweep in E6.
"Two populations is not a tumour."	Named limitation. Strobl's model has fits to 67 patients, and added complexity is not what this paper claims.
"Resistance is pre-existing and irreversible."	Strobl's justification via Viossat and Noble is cited; Ghoreyshi's plasticity work is named as the next step.
"No spatial structure."	Aif showed spatial effects break ODE-trained policies. Named as the principal threat to validity, not hidden.
"No pharmacokinetics."	King 2025 is cited for what that omits, including the adherence result.
"Toxicity is a dose proxy."	R5 uses Gevertz's toxicity ODE rather than cumulative dose, if included.
"Total burden is a surrogate for a surrogate."	West 2023, section 3.2, cited directly.
"gamma = 0.999 over a hundred steps is effectively undiscounted."	Report the effective horizon explicitly and sweep gamma as an ablation.
"No clinical claim is supported."	None is made. The register is Lagzian's: a proof of concept demonstrating feasibility.
"Adaptive therapy itself is not proven."	Mistry's critique is engaged directly, and it is the reason controlled in-silico comparison is the right instrument.

Minimum viable version, and the repository

MVP. Two objectives rather than four: R1 survival against R3 coexistence. That contrast is the sharpest in the literature, since it is Gallagher against Giles & Newton directly, and it is the one a reader will care about. One patient, ten seeds each, cross-evaluation on four

metrics, plus the sigmoid and KL divergence analysis. E0 through E5, nothing else. This answers the research question completely and fits comfortably in eight weeks.

Full version. Four or five objectives, cohort training, parameter perturbation, decision-frequency sweep, and the transfer test. Each addition is a separate figure and none of them changes the core claim, which is what makes the MVP a safe fallback rather than a lesser project.

The repository. Reproducibility is cheap to build in and impossible to retrofit.

```
src/
  env.py          # the ODE, one class, no globals
  objectives.py   # R1-R5, one function each, same signature
  baselines.py    # MTD, AT50, AT-x, AT-PSC, toxicity-feedback
  vi.py           # value iteration per objective
  train.py        # takes a config path, writes a checkpoint
  evaluate.py     # every agent x every metric
configs/          # one YAML per run, no hardcoded numbers
results/          # checkpoints, trajectories, cohort definitions
figures/          # one script per figure, reads results/ only
```

Fix seeds and report them. Generate the virtual cohort from a recorded seed and save it, so the cohort is an artefact rather than a re-roll. Log whole trajectories, not just TTP, because the interpretability figures need S(t), R(t) and D(t) and re-running to recover them is expensive. Pin the environment. Aim for a single command that regenerates every figure from a clean clone.

Four repositories will save days: `MathOnco/AT_costOfResistance_LVModel` for the ODE and fitting code, `Kexin0902/AT-PSC` for the protocol suite and the seven-patient parameter table, `MathOnco/DRL_Personalized_AT` for Gallagher's parameters, and `DavisWeaver/evo_dm` for the MDP and DQN framework.

What not to do

Each of these looks like progress and is not, for a specific reason established in Phase 1.

- **Swap PPO for something else and report it.** Zhu 2026 already compared DDPG, SAC, PPO and A2C on a tumour model, and Kidane's September 2026 preprint compared TD3 and DQN against a Pontryagin benchmark. The algorithm is not the contribution.
- **Add a third or fourth cell population.** Giles already has three, and complexity without a question attached is the specific failure your prompt warns about.

- **Add an LSTM because the problem is partially observable.** Kiram settled this. Doing it again without a new question is replication.
- **Add mutation, immune cells or spatial structure mid-project.** Each changes the environment, which invalidates every completed cross-evaluation run. If you want them, they belong in E10 as a transfer target, never as a mid-flight upgrade.
- **Build the probing-cycle personalisation pathway.** Gallagher proposes it, Lu and Zhang replicate it independently as delayed-I2ADT, and Aif shows ill-timed probing causes premature competitive release. That loop is closed.
- **Report a single best trajectory, or mean TTP alone.** Three separate papers in your corpus show this misleads.
- **Claim clinical applicability.** No patient data, no validation, no claim.
- **Chase a higher number against MTD.** With tuned AT-x and a computable optimum available, beating MTD is table stakes, not a result.

One more, specific to this project: do not let the reward comparison become a hyperparameter comparison. Every objective must be trained with the same algorithm, architecture, step budget and seed set. The moment you tune per objective, the off-diagonal stops meaning anything.

Sources

Everything below was opened and checked. Claims traced to these are marked [A] in the text; inferences across papers are marked [C].

Verified from the PDFs in your corpus. Gallagher et al., *Learning to Adapt*, bioRxiv 2023.04.28.538766 - reward table and the AT50 = 672 d / CT = 451 d figures. Strobl et al., *Cancer Res* 2021;81(4):1135 - r_S = 0.027/day and d-hat_D = 1.5. Giles & Newton, *Phys Rev E* 2026;113:064404 - the entropy reward and the framing quote. Ma et al., *PLOS Comput Biol* 2026;22(6):e1014457. Kiram et al., arXiv:2605.02552v1. Gluzman et al., *Proc R Soc B* 2020;287:20192454 - bang-bang result and sigma = 0.01. Yao et al., *Brief Bioinform* 2024 - the simulate-forward terminal reward.

Verified online.

- Widdershins, Hansen, Read & Hohl, npj Syst Biol Appl 2026;12:27 - authorship confirmed
- Aif et al., bioRxiv 2025.04.08.647768 - authors and the corrected title, "emergent spatial dynamics"

- Lu & Zhang et al., Brief Bioinform 2024;25(2):bbae071 - the I2ADT paper missing from your corpus
- Gevertz et al., npj Syst Biol Appl 2026;12:11 - the toxicity model, also missing from your corpus
- Liu et al., Sci Adv 2025;11:eadv3316
- Kidane, Motayed & Wang, arXiv:2609.12264 - posted 10 September 2026, after your corpus was compiled

Not verified. The corpus's numerical claims for Eastman (0.991 scaled score) and Strobl (Pearson 0.77 cost-turnover correlation) were not checked against their PDFs, and forward-citation search was partial because Google Scholar blocks automated access.